

Chi-Sheng (Michael) Chen (陳麒升)

LLM Agents | Reinforcement Learning | Quantitative Finance | Multi-Modal AI

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Nationality: Taiwan | Open to relocation to Taipei

Chi-Sheng (Michael) Chen is an LLM/RL researcher and engineer with strong quantitative finance experience, holding 45 peer-reviewed and preprint publications (incl. ICASSP×4 with 3 orals, Physical Review A, JAD, JBHI-tier journals), 416+ citations, and **ICML 2026 Gold Reviewer** status. He brings end-to-end LLM/agent capabilities: contributed to RLHF pipelines at **OpenAI** (SFT/DPO, preference modeling, safety alignment); co-founded **Omnis Labs**, operating an AI-driven DeFi liquidity protocol with time-series models running in production (**50–120% base fee APR** on WBTC/USDC and ETH/USDC pairs); won **3rd Place at Berkeley RDI AgentX-AgentBeats** multi-agent competition (1,300+ teams worldwide). His ICASSP 2026 oral on *Quantum Adaptive Self-Attention for Financial Rebalancing* bridges LLM-style attention architectures with real AMM trading.

PROFESSIONAL EXPERIENCE

Omnis Labs

Co-Founder & Lead AI/Quant Engineer

Remote

Sep 2024 – Present

- Built and operate an AI-driven DeFi liquidity positioning protocol —**automated market maker (AMM) strategies, time-series forecasting models, and full backtesting infrastructure.**
- Production results: **50%–120% base fee APR** on WBTC/USDC and ETH/USDC pairs (Katana on SushiSwap) driven by time-series ML strategies; daily execution under real market conditions.
- Built end-to-end MLOps: containerized inference (Docker), feature/model versioning, CI/CD, on-chain event extraction pipeline, and observability dashboards for live trading P&L and slippage attribution.
- Designed LLM-assisted research tooling for market analysis, factor extraction from on-chain and off-chain text feeds, and rapid strategy prototyping.

OpenAI (contractor via Mercor)

AI Trainer —RLHF / Alignment

Remote (US)

Mar 2025 – Oct 2025

- Contributed to **RLHF pipelines**: produced high-quality comparative judgments, preference rankings, and model-behavior assessments used in **SFT, DPO, and reward modeling.**
- Performed high-complexity AI data labeling and evaluation across instruction following, multimodal reasoning, tool use, and safety alignment.
- Drove rubric design and dispute resolution that improved labeling consistency on hard prompts (math reasoning, code, ambiguous policy edges).

Harvard Medical School & Beth Israel Deaconess Medical Center

Research Affiliate —Advisor: Prof. Gabriel Brat

MA, USA

Nov 2024 – Present

- Built real-time EMS triage **multi-agent** pipeline: ASR (Whisper) → event extraction with LLM agents → structured trauma prediction. Same agentic patterns (RAG, tool use, task planning) directly transfer to trading research workflows.
- Designed prompt engineering and fine-tuning pipelines (LoRA/SFT) over domain-specific text corpora for clinical event extraction.

Neuro Industry, Inc.

Co-Founder & CTO

CA, USA

Mar 2024 – Jan 2025

- Built GCP-based MLOps pipeline (Vertex AI, Cloud Run, BigQuery, Cloud Build); enabled **end-to-end training on 10K+ samples** with reproducible model versioning.
- Shipped 3 research products on quantum ML, generative diffusion, and EEG foundation models.
- Owned hiring, technical architecture, cloud cost optimization, and investor technical due diligence.

Department of Computer Science, National Yang Ming Chiao Tung University

Research Assistant —Advisor: Prof. Chun-Shu Wei

Hsinchu, Taiwan

Dec 2023 – Aug 2024

- Developed **MUSE**, a multimodal contrastive learning framework (28 citations) using techniques transferable to text–time-series alignment (e.g., news ↔ price action).

MediaTek Inc.

Digital IC R&D Engineer

Hsinchu, Taiwan

Dec 2021 – Oct 2023

- Microprocessor IP development for flagship 5G smartphones' display and AMBA SoC —strong systems-level foundation for HPC/GPU/FPGA latency-sensitive engineering.
- Hardware virtualization architecture RTL design and IP verification (UVM, SystemVerilog) —disciplined spec-driven and test-driven workflow.

SELECTED PROJECTS (LLM / AGENTS / QUANT FINANCE)

- MadGAA — OSCE Medical Multi-Agent Judge System** (*Berkeley AgentX-AgentBeats, 3rd Place / 1,300+ teams*) 2026
- Co-designed a multi-agent LLM evaluation system with role-specialized agents (examiner, patient, judge) and a verifiable scoring rubric.
 - Same agentic architecture (RAG + tool use + verifier) is directly portable to trade-strategy review, signal verification, and execution oversight.
 - Repo: github.com/MadGAA-Lab/OSCE-Project
- QASA — Quantum Adaptive Self-Attention for Financial Rebalancing** (*ICASSP 2026, oral*) 2026
- Empirical study of hybrid quantum-classical attention applied to AMM rebalancing in decentralized finance; backtested across multi-asset universes with rigorous P&L attribution.
- QRL Sector Rotation Trading Agent** (*IEEE QCE 2025*) 2025
- Reinforcement learning agent for sector rotation on Taiwan stock market; closed-loop training with realistic execution costs.
- Large Cognition Model — Pretrained EEG Foundation Model** (*arXiv 2502.17464*) 2025
- End-to-end self-supervised pretraining of a foundation model over heterogeneous time-series; same recipe (mask-prediction + contrastive) generalizes to financial time-series pretraining.
- NYCU EP Laser Physics AI Tutor** (*RAG-based educational LLM, in production*) 2025–Present
- Next.js + Gemini + Supabase pgvector RAG system with 8 learning modes (adaptive quiz, exam simulation, knowledge graph, spaced repetition). Serving real NYCU students. Live demo: web-eight-hazel-22.vercel.app.

SKILLS

LLM Stack: PyTorch, HuggingFace Transformers/PEFT, LoRA, QLoRA, SFT, DPO, RLHF, quantization (GPTQ/AWQ), distillation, MoE, prompt engineering, vLLM, OpenAI SDK, Anthropic SDK, Gemini SDK, LangChain, LangGraph, Ollama, RAG (pgvector, FAISS, Qdrant)

Agent / Tooling: Tool use, function calling, ReAct, multi-agent orchestration, Cursor, Claude Code, GitHub Copilot, TDD / spec-driven development

ML/RL: Reinforcement learning (PPO/DPO/Q-learning), contrastive learning, diffusion models, transformers, state-space models, time-series forecasting, Scikit-learn, MLflow, Weights & Biases

MLOps / Infra: Docker, CI/CD (GitHub Actions), Kubernetes basics, GCP (Vertex AI, BigQuery), AWS, model/feature versioning, observability (Prometheus, Grafana)

Quant / Finance: Backtesting, factor modeling, event extraction from financial text, AMM/DEX strategy design, on-chain data pipelines, Solidity (read-only contracts), Python (pandas, NumPy, statsmodels)

Systems: Python, C/C++, Rust, JavaScript/TypeScript, Next.js, MySQL/PostgreSQL, Git, Verilog/SystemVerilog (legacy)

Languages: Chinese (Native), English (Fluent —1.5+ yrs working in US research environments)

SELECTED PUBLICATIONS (45 PAPERS, 416+ CITES, H-INDEX 12, I10-INDEX 16)

[LLM / Agents / Quant Finance / RL]

- Chen, C.-S., Tsai, A.H.-W. “Quantum Adaptive Self-Attention for Financial Rebalancing: An Empirical Study on Automated Market Makers in Decentralized Finance” ICASSP 2026 (oral), doi: 10.1109/ICASSP55912.2026.11460972.
- Chen, C.-S., Tsai, A.H.-W. “Quantum and Classical Machine Learning in Decentralized Finance: Multi-Asset Backtesting of Automated Market Makers” arXiv:2510.15903 (2025).
- Chen, C.-S., Tsai, A.H.-W. “Benchmarking Classical and Quantum Models for DeFi Yield Prediction on Curve Finance” arXiv:2508.02685 (2025).
- Chen, C.-S., Zhang, X., Chen, Y.-C. “Quantum Reinforcement Learning Trading Agent for Sector Rotation in the Taiwan Stock Market” IEEE QCE 2025, arXiv:2506.20930.
- Chen, C.-S., Zhang, X., Kuo, E.-J., et al. “Quantum Temporal Convolutional Neural Networks for Cross-Sectional Equity Return Prediction” arXiv:2512.06630 (2025).
- Chen, C.-S., Kuo, E.-J. “Quantum-Enhanced Natural Language Generation: A Multi-Model Framework with Hybrid Quantum-Classical Architectures” arXiv:2508.21332 (2025).
- Chen, C.-S., Kuo, E.-J. “Quantum Adaptive Self-Attention for Quantum Transformer Models” arXiv:2504.05336 (2025).
- Chen, C.-S., Chen, Y.-J., Tsai, A.H.-W. “Large Cognition Model: Towards Pretrained EEG Foundation Model” arXiv:2502.17464 (2025).
- Chen, C.-S., Kuo, E.-J. “Do Quantum Transformers Help? A Systematic VQC Architecture Comparison on Tabular Benchmarks” arXiv:2604.23931 (2026).
- Chen, Y.-J., Albarqawi, A., Chen, C.-S. “Enhancing Clinical Decision-Making: Integrating Multi-Agent Systems with Ethical AI Governance.” IEEE CIBCB 2025, doi: 10.1109/CIBCB66090.2025.11177136.
- Lu, J.T.Y., Chiang, J., Chen, C.-S., et al. “EEG2text-CN: Open-Vocabulary Chinese Text-EEG Alignment via LLM and Contrastive Learning.” arXiv:2506.00854 (2025).

[Selected Peer-Reviewed — Other Top Venues]

Chen, C.-S., Kuo, E.-J. “Quantum Reinforcement Learning-Guided Diffusion Model for Image Synthesis” ICASSP 2026 (oral), doi: 10.1109/ICASSP55912.2026.11461991.

Chen, C.-S., Tsai, A.H.-W., Huang, S.-C. “Quantum Multimodal Contrastive Learning Framework” ICASSP 2025 (oral), doi: 10.1109/ICASSP49660.2025.10889504.

Chen, C.-S., Kuo, E.-J. “Unraveling Quantum Environments: Transformer-Assisted Learning in Lindblad Dynamics” Physical Review A (2025).

Chen, C.-S., et al. “QEEGNet: Quantum Machine Learning for Enhanced EEG Encoding” IEEE SiPS 2024 / Journal of Signal Processing Systems (2025).

Li, C.-T., Chen, C.-S., et al. “Prediction of Antidepressant Responses to Non-Invasive Brain Stimulation Using Frontal EEG.” Journal of Affective Disorders (2023), IF=6.5.

EDUCATION

National Taiwan University (NTU) — College of EECS Taipei, Taiwan
M.S., Biomedical Electronics & Bioinformatics — GPA: 3.74/4.30 Sep 2019 – Jun 2021
Advisors: Prof. Cheng-Ta Li, MD & Prof. Chung-Ping Chen

- Thesis: Attention-based classical/quantum ML for non-stationary time-series prediction.
- Resulting EEG model deployed at the Precision Depression Intervention Center, Taipei Veterans General Hospital.

National Chiao Tung University (NCTU) Hsinchu, Taiwan
B.Eng., Electrophysics / B.S., Interdisciplinary Science Degree Program Sep 2015 – Jun 2019

- Triple background: Electrical Engineering, Applied Mathematics, Life Sciences — direct foundation for quantitative cross-disciplinary work.

PROFESSIONAL SERVICE

ICML 2026 Gold Reviewer — recognized among the top reviewers based on area chair ratings.

Conference Reviewer (2026): NeurIPS, ICML, KDD, MICCAI, ICASSP.

Program Committee: QNRL Workshop @ IEEE WCCI 2026.

IOP Trusted Reviewer (2026) — Institute of Physics Publishing.

Journal Reviewer (selected): TPAMI, TNNLS, TASLP, JBHI, IEEE TNSRE, npj Quantum Information, Oxford National Science Review, IOP Journal of Neural Engineering.

AWARDS & ACHIEVEMENTS

2026 Berkeley RDI AgentX-AgentBeats: Phase 1 Healthcare Multi-Agent System — **3rd Place** (Team MadGAA, 1,300+ teams worldwide).

2026 NVIDIA GTC Top 8 Finalist: “GPU-Accelerated Chinese Brain-to-Text Interface Powered by NVIDIA H100 80GB.”

2026 ICML Gold Reviewer.

2023 Certificate of the 20th National Innovation Award — clinical research category.

2019 International Blockchain Olympiad (IBCOL): World Finalist (100+ teams).

2017 Microsoft International Imagine Cup: 2nd place in Taiwan.

2016 International Genetically Engineered Machine Competition (iGEM): World Gold medal, Best Applied Design, Best Part Collection, Best Presentation (300+ university teams).